SDU University
Faculty of Engineering and Natural Science

Diploma Project

Development of an Intelligent Digital Healthcare Platform for Data-Driven Medical Decision Support

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6B06102 - ”Information Systems”

Kaskelen, 2026

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Development of an Intelligent Digital Healthcare Platform for Data-Driven Medical Decision Support

A thesis submitted for the degree of
Bachelor in Computer Science
(degree code: 6B06102)

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Abstract

The rapid advancement of digital technologies and Artificial Intelligence (AI) has remarkably changed the healthcare realm. Our project's main objective is to collaborate medical decision-making with AI. Moreover, it assists doctors in diagnosing diseases and analyzing patients' data.

The proposed platform combines medical databases, Large Language Models (LLM), and chat-bots for evaluating information of laboratory and clinical indicators. By operating these technologies, the system helps to anticipate potential illnesses and recommend applicable treatment plans. The architecture of the project is constructed with emphasis on data integrity, scalability, and integration with the Kazakhstani healthcare system. Moreover, the platform is suggested with a high user-friendly interface for better interaction. The implementation of that website facilitates to diminish human error, increase the accuracy of medical diagnostic, and embellish overall clinicians' outcomes. Additionally, it will contain better personalized medical care of sick people tailored on its own patient data.

To conclude, that AI based intelligent healthcare platform shows its impact on modern medicine and highlights its features for data-driven decisions.

Index terms: Artificial Intelligence(AI) in healthcare, Large Language Model (LLM model), data analytics, intelligent digital healthcare platform , decision-making.

Аңдатпа

Сандық технологиялар мен жасанды интеллекттің (ЖИ) қарқынды дамуы денсаулық сақтау саласын айтарлықтай өзгертті. Біздің жобаның негізгі мақсаты - медициналық шешім қабылдауда ЖИ-мен бірлесіп жұмыс істеу. Сонымен қатар, ол дәрігерлерге ауруларды анықтауға және науқастардың деректерін талдауға көмектеседі.

Ұсынылған платформа медициналық деректер қорын, үлкен тілдік модельдерді (LLM), зертханалық және клиникалық көрсеткіштер туралы ақпаратты бағалауға арналған чат-боттарды біріктіреді. Осы технологияларды пайдалану арқылы жүйе ықтимал ауруларды болжауға және қолданылатын емдеу жоспарларын ұсынуға көмектеседі. Жобаның архитектурасы деректердің тұтастығына, ауқымдылығына және қазақстандық денсаулық сақтау жүйесінің интеграциясына баса назар аудара отырып жасалған. Сонымен қатар, платформа пайдаланушыларға өзара әрекеттесу үшін ыңғайлы интерфейспен ұсынылады. Бұл веб-сайтты енгізу адам қатесін азайтуға, медициналық диагностиканың дәлдігін арттыруға және жалпы клиникалық нәтижелерді безендіруге көмектеседі. Оған қоса, ол пациенттердің жеке деректеріне сәйкес науқастарға жекелендірілген медициналық көмекті қамтиды.

Қорытындылай келе, AI негізіндегі зияткерлік денсаулық сақтау платформасы заманауи медицинаға әсерін көрсетеді және деректерге негізделген шешімдер үшін оның мүмкіндіктерін көрсетеді.

Аннотация

Быстрое развитие цифровых технологий и искусственного интеллекта (ИИ) значительно изменило сферу здравоохранения. Главная цель нашего проекта — объединить принятие медицинских решений с ИИ. Более того, это помогает врачам в диагностике заболеваний и анализе данных пациентов. Предложенная платформа объединяет медицинские базы данных, большие языковые модели (LLM) и чат-боты для оценки информации о лабораторных и клинических показателях. Используя эти технологии, система помогает прогнозировать потенциальные заболевания и рекомендовать соответствующие планы лечения. Архитектура проекта построена с акцентом на целостность данных, масштабируемость и интеграцию с системой здравоохранения Казахстана. Кроме того, платформа предлагается с высоко удобным пользовательским интерфейсом для лучшего взаимодействия.

Внедрение этого веб-сайта способствует уменьшению человеческих ошибок, повышению точности медицинской диагностики и улучшению общих результатов работы врачей. Кроме того, он будет обеспечивать более персонализированную медицинскую помощь больным, адаптированную к собственным данным пациентов. В заключение, эта интеллектуальная платформа здравоохранения на основе ИИ демонстрирует свое влияние на современную медицину и подчеркивает свои возможности для принятия решений на основе данных.

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1 Introduction

1.1 Problem Background

Healthcare systems are producing a large amount of information from different types of electronic devices and wearable devices. Despite the availability, many health services lack resources for precise decision-making. One of the problems is the complexity and volume of medical data. Skilled people in healthcare must process enormous amounts of information under pressure. Without Artificial Intelligence(AI) tools, this boosts the likelihood of errors and cognitive overload. The shortage of AI and developed analytics in routine activities results in improper patient outcomes. Currently, majority systems cannot provide tailored recommendations based on individual data. As a result, there is a demand for a digital healthcare platform that can merge miscellaneous medical information and applies different techniques.

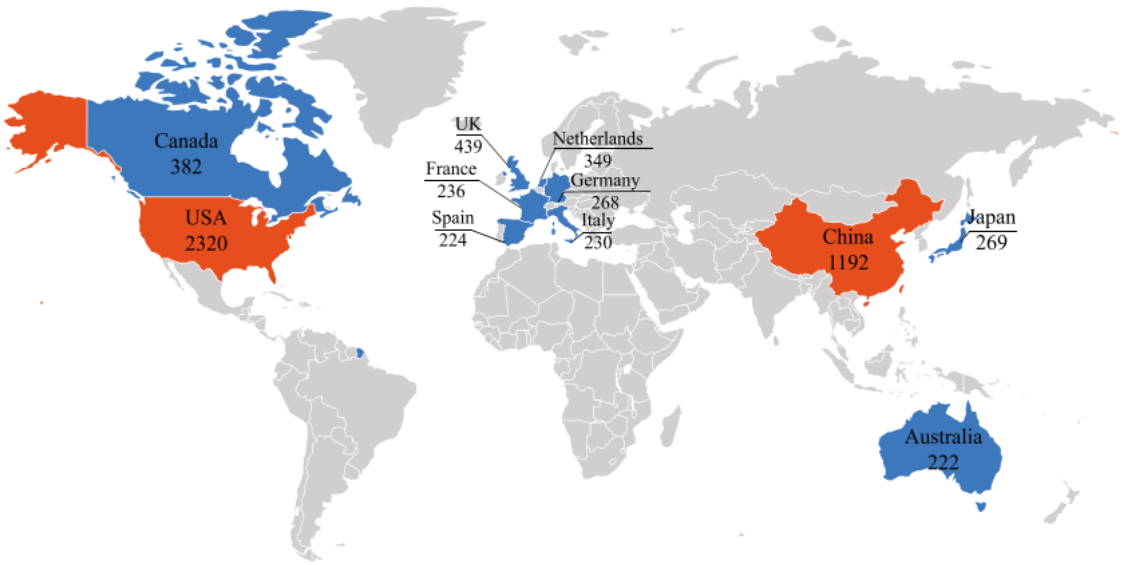


FIGURE 1. Top ten countries publishing papers regarding data-driven smart healthcare systems.

[3]

1.2 Aim of the study and objectives

The aim of the project is create AI based platform that will assist doctors for decision-making by data analytics. It will include couple of objectives for achieving aim. There are:

1. Inspect existing digital healthcare systems and detect their limitations.
2. Design the architecture of the website and guarantee its interconnectivity and competent data management.
3. Implement data analytics, AI, and Large Language Model (LLM) model for supporting decision-making.
4. Develop user-friendly interface for better interaction among users.
5. Based on the results of analysis, give suggestions for improving it.

1.3 Significance of the Study

The significance of this study lies in its contribution to improving the quality and efficiency of healthcare services through the application of intelligent digital technologies. By developing a data-driven medical decision support platform, this project addresses the growing need for more accurate, timely, and evidence-based clinical decisions. First, the proposed system can enhance clinical decision-making by providing healthcare professionals with real-time insights and analytical support based on patient data. This reduces the likelihood of human error and improves diagnostic accuracy, especially in complex medical cases. Second, the study contributes to the integration and effective use of medical data. By combining data from multiple sources into a unified platform, it helps overcome the problem of data fragmentation and enables a more comprehensive view of patient health.

Finally, this study has practical and academic value. Practically, it offers a foundation for future implementation of intelligent healthcare systems in medical institutions. Academically, it contributes to the fields of digital healthcare, data analytics, and medical informatics, providing a basis for further research and development.

1.4 Scope and Limitations

This project focuses on the development of an intelligent digital healthcare platform designed to support medical decision-making through data-driven approaches. The system includes the following functionalities:

1. Integration of a large language model (LLM) to provide intelligent responses and assist users in understanding medical information.

2. Implementation of a chatbot system that interacts with users, answers health-related questions, and provides basic medical guidance.

3. A pharmacy directory feature, allowing users to access information about available pharmacies.

4. Support for multiple language options, improving accessibility for diverse users.

5. Basic healthcare platform functionalities, including navigation, user interaction, and data presentation.

6. A recommendation system that provides suggestions based on user input and analysis results.

Additionally, the platform includes an informational module related to human anatomy, providing users with visual and educational content. Despite its functionality, the system has several limitations that require further improvement:

1. The human anatomy module is currently limited in scope. At present, it only displays the skeletal structure and muscular system of male and female bodies. More detailed anatomical systems and interactive features are needed for a comprehensive learning experience.

2. The AI assistant, although functional, requires further training and optimization. Its performance can be significantly improved by incorporating a larger and more specialized dataset, particularly focused on healthcare information relevant to the Republic of Kazakhstan.

3. The platform currently provides basic-level medical support and should not be considered a replacement for professional medical consultation.

Overall, while the system demonstrates the potential of intelligent healthcare platforms, further development and enhancement are necessary to expand its capabilities, improve accuracy, and ensure broader applicability in real-world healthcare environments.

2 Literature Review

That part includes theoretical part of our diplomic project.

2.1 Digital Healthcare Systems

Digital healthcare systems refer to the integration of information and communication technologies (ICT) into healthcare services to improve accessibility, efficiency, and quality of medical care. These systems include electronic health records (EHR), telemedicine, mobile health applications, and hospital information systems.

Globally, digital healthcare has become a key trend due to the increasing volume of medical data and the need for efficient data management. In the Republic of Kazakhstan, digital transformation of healthcare has been actively developing over the past decades. Initial steps began in the 1990s with the introduction of basic information systems for statistical data collection, followed by the creation of disease registries in the early 2000s .

Currently, Kazakhstan is implementing a national e-health system, including electronic medical records, online appointment systems, and mobile healthcare applications. More than 2 million users (approximately 17% of the adult population) actively use digital medical applications, which allow access to personal health data and medical services . Furthermore, over 87% of public healthcare services have been digitalized, demonstrating a high level of system automation.

2.2 Clinical Decision Support Systems (CDSS)

Clinical Decision Support Systems (CDSS) are computer-based tools designed to assist healthcare professionals in making clinical decisions. These systems analyze medical data and provide recommendations, alerts, or diagnostic support.

CDSS can be categorized into several types:

1. Knowledge-based systems (rule-based)
2. Machine learning-based systems
3. Predictive analytics systems

The main purpose of CDSS is to reduce medical errors, improve diagnostic accuracy, and enhance treatment outcomes. In digital healthcare environments, CDSS are often integrated with electronic health records, enabling real-time data analysis.

In Kazakhstan, the development of CDSS is closely linked with the national digital healthcare strategy, which emphasizes improving the quality and efficiency of medical services through data-driven approaches . The transition to paperless healthcare and unified data systems creates a foundation for implementing intelligent decision support tools.

However, many existing systems still lack advanced analytics and real-time decision support, highlighting the need for more intelligent platforms.

2.3 Artificial Intelligence in Healthcare

Artificial Intelligence (AI) has become a transformative technology in healthcare, enabling automated data analysis, predictive modeling, and personalized medicine. AI techniques such as machine learning, deep learning, and natural language processing are widely used in diagnostics, medical imaging, and clinical decision support.

One of the most recent advancements in AI is the development of Large Language Models (LLMs), which can process and generate human-like text. In healthcare, LLMs can be used for:

1. Patient communication
2. Medical information retrieval
3. Clinical documentation
4. Decision support

In Kazakhstan, the integration of AI into healthcare is still developing but is supported by national digitalization strategies. The government emphasizes the importance of data processing, storage, and protection, as well as the development of digital infrastructure for healthcare systems. For instance, edensaulyq is the digital platform.

Despite the potential benefits, challenges remain, including data availability, model accuracy, and the need for localization of AI systems using region-specific medical data.

2.4 Chatbots in Healthcare

Healthcare chatbots are AI-powered systems that interact with users through text or voice interfaces. They are increasingly used to provide medical information, symptom checking, appointment scheduling, and patient support.

The advantages of healthcare chatbots include:

1. 24/7 availability
2. Reduced workload for healthcare professionals
3. Immediate responses to patient queries
4. Improved accessibility to medical information

In the context of Kazakhstan, chatbot technologies can significantly enhance access to healthcare services, especially in rural and remote areas where medical resources are limited. The government's focus on telemedicine and digital services supports the adoption of such technologies. However, chatbots also have limitations, such as limited medical accuracy and the inability to replace professional medical consultation.

3 Methodology

That part includes core technical part of our diplomic project.

3.1 System Overview

This study focuses on the development of an intelligent digital healthcare platform designed to support medical decision-making through data-driven approaches. The system integrates artificial intelligence technologies, including a large language model (LLM), with user-centered functionalities such as a chatbot, recommendation system, and healthcare information modules. The platform is designed to assist users in accessing medical information, receiving basic health guidance, and navigating healthcare services such as pharmacies. It provides an interactive environment where users can input queries and receive real-time responses, recommendations, and relevant healthcare data.

In the context of the Republic of Kazakhstan, where digital healthcare adoption is increasing, such platforms contribute to improving accessibility and efficiency of medical services, especially in regions with limited healthcare infrastructure.

3.2 System Architecture

The architecture of the proposed system follows a modular and scalable design, consisting of several interconnected components:

1. User Interface Layer — responsible for interaction between the user and the system, including web pages, chatbot interface, and visualization elements.
2. Application Layer — handles business logic, user requests, and system operations.
3. AI Layer — includes the large language model and recommendation engine, which process user inputs and generate responses.
4. Data Layer — manages storage and retrieval of healthcare data, including pharmacy information and system datasets.

The architecture ensures separation of concerns, enabling efficient system development and future scalability. Data flows from the user interface to the backend, where it is processed and analyzed before generating a response. This modular design aligns with modern healthcare system architectures and supports integration with external data sources in future developments.

3.3 Data Sources and Processing

The system utilizes multiple types of data to support its functionality:

1. User input data — queries entered by users through the chatbot interface.
2. Healthcare-related datasets — including information about pharmacies and basic medical knowledge.
3. Pre-trained language model data — used by the LLM for generating responses.

Data processing involves:

1. Input preprocessing (text cleaning and formatting)
2. Query interpretation
3. Response generation using AI models
4. Output formatting for user display

In Kazakhstan, healthcare data is increasingly digitized through national e-health systems. However, access to structured and publicly available datasets remains limited. Therefore, the system primarily relies on general medical knowledge and locally available data sources. Future improvements may include integration with national healthcare databases to enhance accuracy and localization.

3.4 LLM Integration

A key component of the system is the integration of a large language model (LLM), which enables natural language interaction and intelligent response generation.

The LLM processes user queries and generates context-aware responses based on:

1. Pre-trained knowledge
2. Prompt design and system instructions
3. Contextual understanding of user input

Although the LLM performs effectively, its accuracy depends on the quality and relevance of training data. In the context of Kazakhstan, further improvement requires incorporating region-specific healthcare information, including local medical guidelines and language adaptation.

3.5 Chatbot Design

The chatbot serves as the primary interaction mechanism between users and the system. It is designed to simulate human-like conversation and provide intuitive access to healthcare information. The chatbot workflow consists of that principles:

1. User submits a query
2. System processes the input
3. LLM generates a response
4. System returns the response to the user

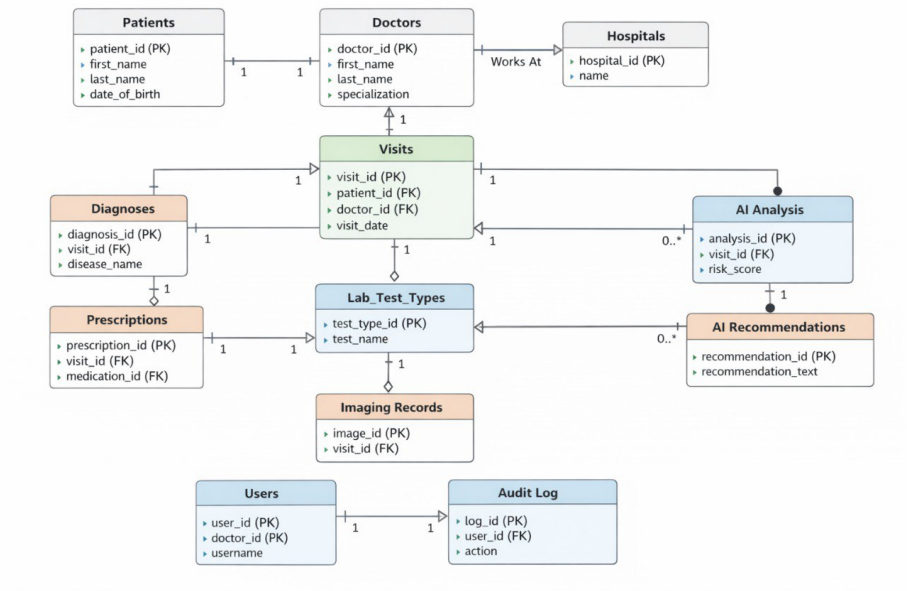
The chatbot supports several functions. For instance, general health inquiries, navigation assistance within the platform, and basic recommendations. The design emphasizes usability, responsiveness, and clarity of communication. It allows users with different levels of digital literacy to interact with the system effectively.

3.6 Recommendation System

The recommendation system provides users with suggestions based on their input and system analysis. It enhances the functionality of the platform by offering personalized or context-based guidance. Recommendations may include: general health advice, suggested actions based on symptoms, and relevant healthcare information.

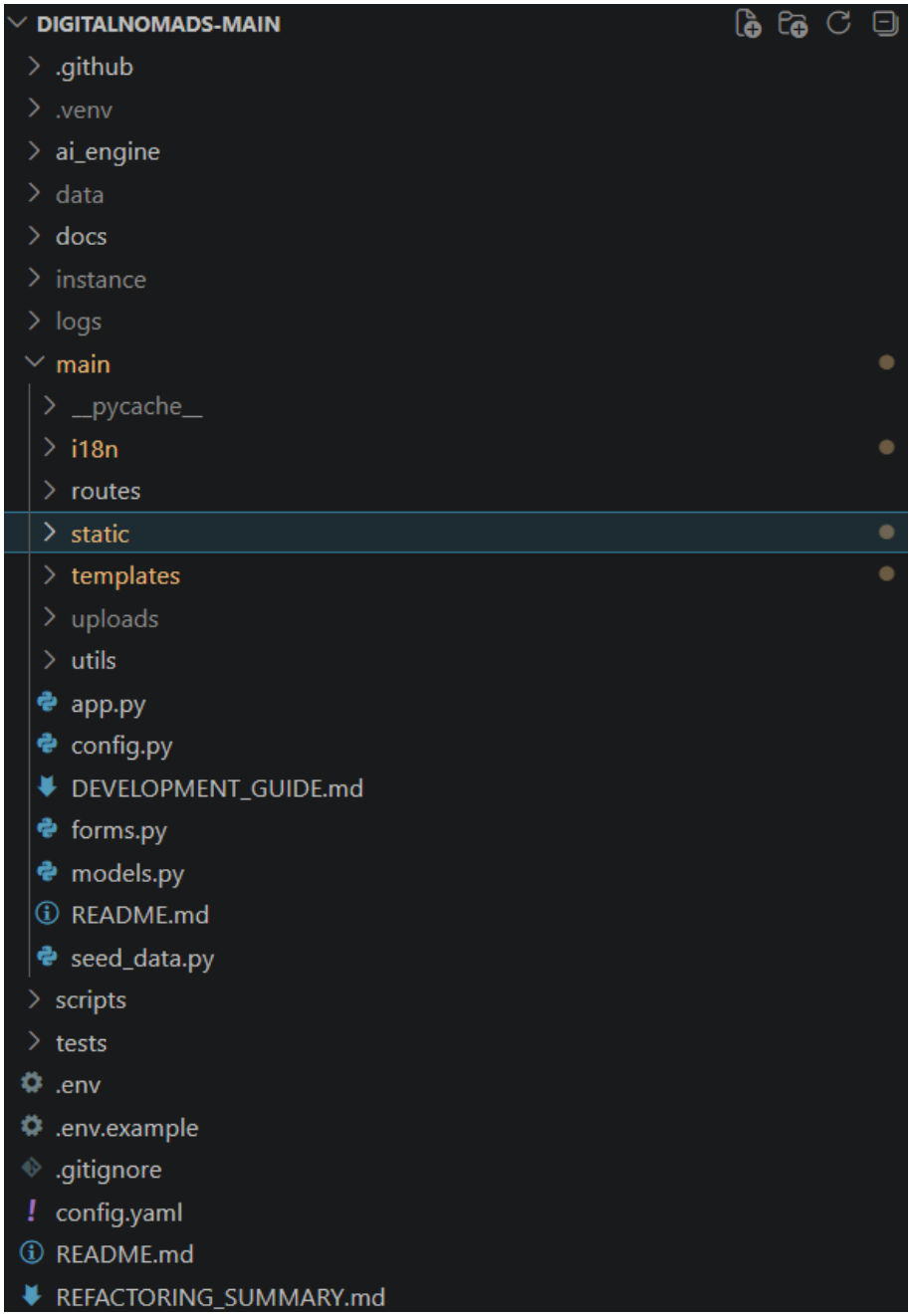
The system uses rule-based logic combined with AI-generated insights. While the current implementation provides basic recommendations, future enhancements may include advanced machine learning models for predictive analytics.

3.7 Database Architecture



3.8 Technology Stack

4 Implementation



The system architecture is designed as a modular multi-layer structure consisting of user interface, application logic, artificial intelligence components, and data management layers.

The user interacts with the system through a web-based interface, which sends requests to the backend server implemented using a Python-based framework. The backend processes user input and distributes tasks among different system components.

The artificial intelligence layer includes a large language model (LLM), which generates natural language responses, and a recommendation module, which provides context-based suggestions. The data layer stores structured information such as pharmacy data and supports efficient data retrieval.

The system components communicate through well-defined interfaces, ensuring scalability, flexibility, and ease of integration with future healthcare data sources.

5 Conclusion and Future Work

5.1 Conclusion

This study focused on the development of an intelligent digital healthcare platform designed to support medical decision-making through data-driven approaches. The primary objective was to integrate modern technologies, including artificial intelligence and data analytics, into a user-friendly system that improves access to healthcare information and assists users in making informed decisions.

The developed platform successfully combines several key components, including a large language model (LLM), a chatbot interface, a recommendation system, and a pharmacy information module. These features enable users to interact with the system in natural language, receive relevant medical information, and access healthcare-related services efficiently.

The implementation of the chatbot demonstrates the effectiveness of conversational interfaces in healthcare applications. Users can ask questions and receive real-time responses, which enhances accessibility and reduces the need for direct interaction with healthcare professionals for basic inquiries. Additionally, the recommendation system provides context-based suggestions, contributing to a more personalized user experience.

The platform also includes support for multiple languages, which increases its usability in diverse populations. The integration of pharmacy data further enhances the practical value of the system by helping users locate relevant healthcare services.

5.2 Future Work

Despite the successful implementation of the core functionalities, several areas for future improvement and development have been identified.

First, the accuracy and reliability of the AI assistant can be enhanced by further training the large language model using more specialized and localized datasets. In particular, incorporating healthcare data relevant to the Republic of Kazakhstan, including national medical guidelines and terminology, would significantly improve the quality and relevance of responses.

Second, the human anatomy module requires further development. Currently, it provides limited visualization of the skeletal and muscular systems for male and female bodies. Future work should expand this module to include additional anatomical systems, such as the nervous, cardiovascular, and digestive systems, as well as interactive and educational features.

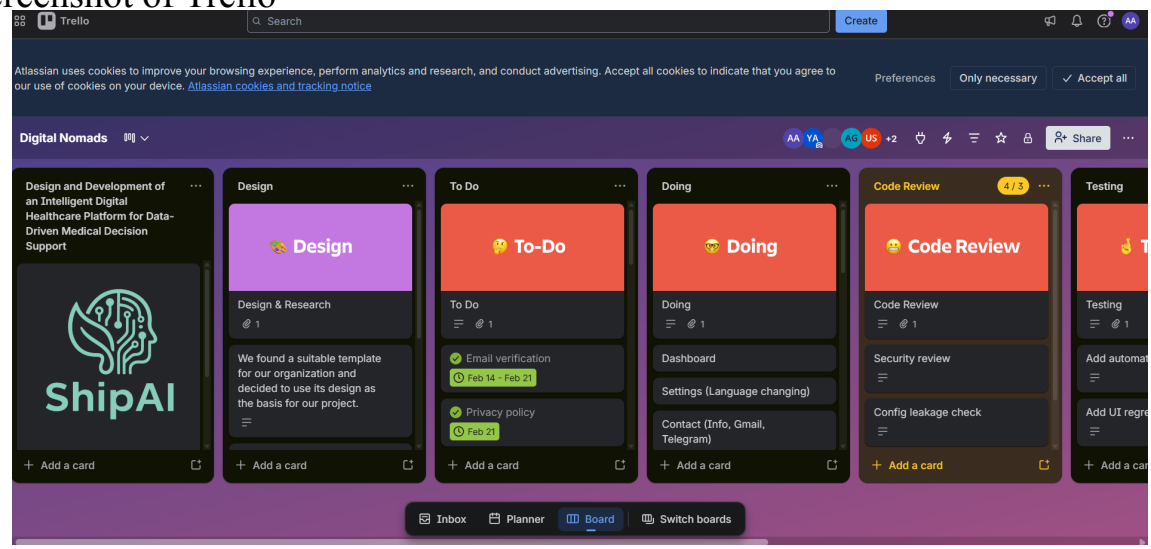
Third, the recommendation system can be improved by implementing more advanced machine learning techniques. This would allow for more accurate predictions, personalized recommendations, and risk assessment based on user input and historical data.

Additionally, future versions of the platform may include integration with real healthcare systems, such as hospitals, electronic health records, and national healthcare databases. This would enable real-time data access and significantly enhance the practical applicability of the system. Another important direction is improving data security and compliance with healthcare regulations. As the system evolves to handle more sensitive data, it will be essential to ensure full compliance with legal and ethical standards.

Finally, the user interface can be further refined to improve usability and accessibility, particularly for users with limited technical skills.

6 Appendix A

Screenshot of Trello



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